

Create and Configure a vNet in Azure (SOP)

Task: create and configure a Virtual Network (vNet) and Subnet in Microsoft Azure

- Create a new vNet, configuring address spaces, and setting up subnets to meet the needs of different applications.
- Apply network security features like encryption and use tags to categorize your vNet.

Virtual networks are important for organizing & managing network traffic in the Azure environment

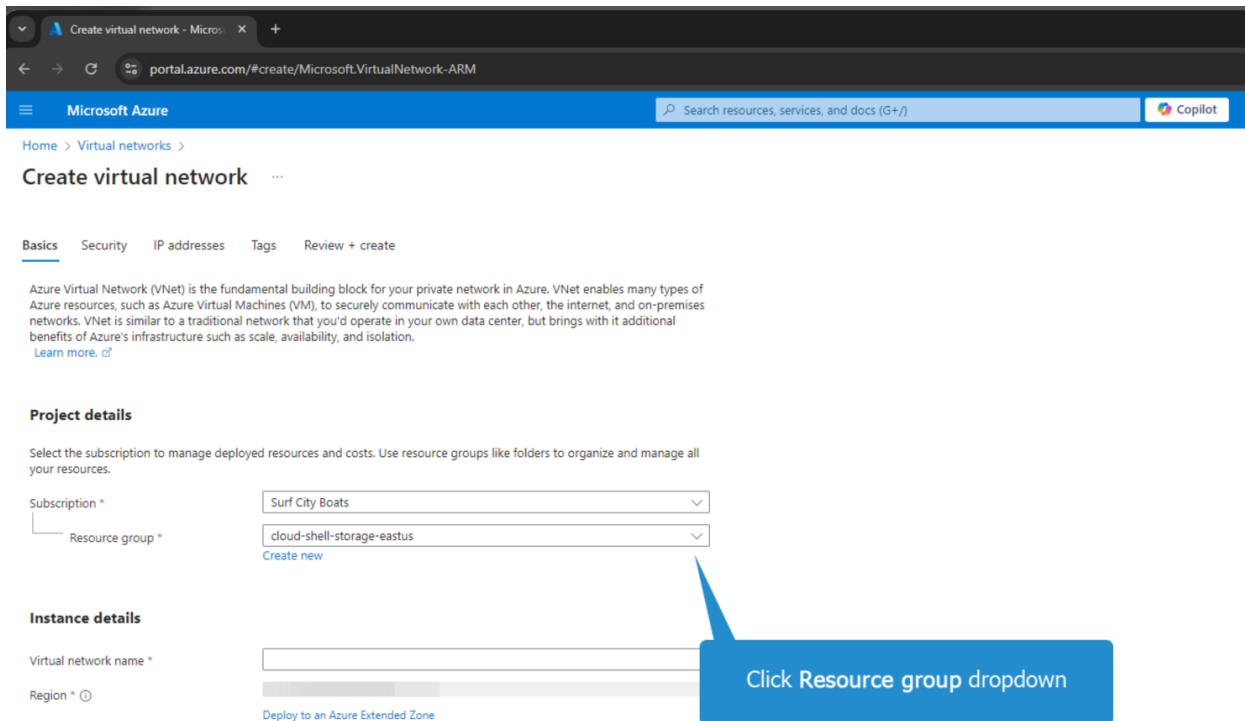
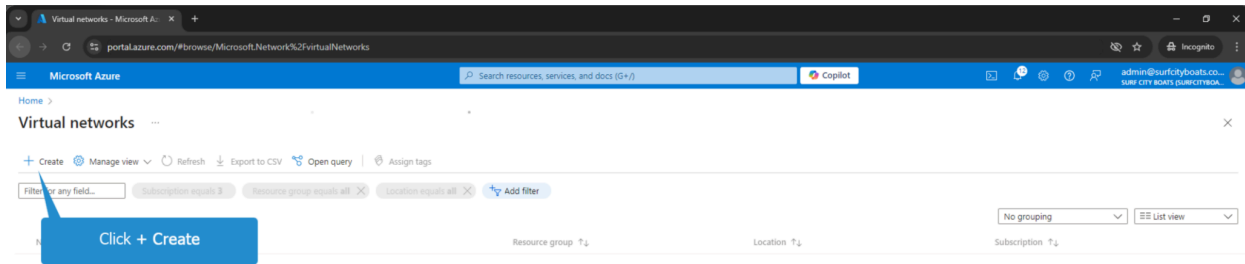
- Subnets segment your vNet to help control resource communication

Task information

- vNet Name: vNet-App in resource group Prod-RG
- Address Space: 192.168.0.0
- Subnet Name: apps
 - Subnet Purpose: Default
- Encryption: Virtual network encryption enabled
- Tag: Environment = Production

The screenshot shows the Microsoft Azure portal interface. At the top, there's a search bar and a 'Copilot' button. Below that, the 'Azure services' section displays various service icons, with 'Virtual networks' highlighted by a blue callout box containing the text 'Click Virtual networks'. The 'Resources' section is visible below, showing a table of existing resources.

Name	Type	Updated
ProdRG	Resource group	a few seconds ago
vmss01	Virtual machine scale set	2 hours ago
scbregistry123	Container registry	2 hours ago
test	Resource group	2 hours ago
SRV01	Virtual machine	5 hours ago
scbstorage12345	Storage account	4 days ago
vNet-Prod	Virtual network	4 days ago
DC01-vnet	Virtual network	4 days ago
DC01	Virtual machine	5 days ago
SurfSide Boats	Subscription	2 weeks ago
Production	Resource group	2 weeks ago
SCBDC1	Virtual machine	2 weeks ago



Create virtual network - Micros

portal.azure.com/#create/Microsoft.VirtualNetwork-ARM

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

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Create virtual network

[Basics](#) [Security](#) [IP addresses](#) [Tags](#) [Review + create](#)

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation.

[Learn more.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Surf City Boats

Resource group *

ProdRG

[Create new](#)

Instance details

Virtual network name *

vNet-App

Region * ⓘ

(US) East US

[Deploy to an Azure Extended Zone](#)

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Home > Virtual networks >

Create virtual network

BasicsSecurityIP addressesTagsReview + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

Add IPv4 address space

192.168.0.0/16

192.168.0.0 /16

192.168.0.0 - 192.168.255.255

65,536 addresses

Delete address space

+ Add a subnet

Subnets	IP address range	Size	NAT gateway
default	192.168.0.0 - 192.168.0.255	/24 (256 addresses)	-

PreviousNextReview + create

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purposeDefault

Namedefault2

IPv4

Include an IPv4 address space☒

IPv4 address range192.168.0.0/16

Starting address192.168.0.0

Size/24 (256 addresses)

Subnet address range192.168.1.0 - 192.168.1.255

IPv6

Include an IPv6 address space☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Enable private subnet (no default outbound access)☒

Security

Simplify internet access for virtual machines by using a network address translation gateway. Filter subnet traffic using a network security group. [Learn more](#)

NAT gatewayNone

Create new

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

AddCancel

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Nameapps

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AddCancel

portal.azure.com/#Create/Microsoft.VirtualNetwork-ARM

Microsoft Azure

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192.168.0.0 /16

192.168.0.0 - 192.168.255.255

65,536 addresses

Delete address space

Add a subnet

Subnets	IP address range	Size	NAT gateway
default	192.168.0.0 - 192.168.0.255	/24 (256 addresses)	-

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Add a subnet

NAT gateway

None

Create new

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

When operating on a private subnet, certain functionalities such as Windows Activation and host operating system updates may not be available without outbound access. To enable outbound connectivity for virtual machines in this subnet, it is recommended to select a NAT gateway

Network security group

None

Create new

Route table

None

Service Endpoints

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#)

Services

Remove service endpoint

Select a service endpoint

Subnet Delegation

Delegation

Click Add

Network policy for Private endpoints

The network policy affects the types of network policies that control traffic going to the private endpoints in this subnet. [Learn more](#)

Private endpoint network policy

Disabled

Add

Cancel

Home > Virtual networks >

Create virtual network ...

Basics Security **IP addresses** Tags Review + create

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Add IPv4 address space

192.168.0.0/16 [Delete address space](#)

192.168.0.0 /16

192.168.0.0 - 192.168.255.255 65,536 addresses

+ Add a subnet

Subnets	IP address range	Size	NAT gateway
default	192.168.0.0 - 192.168.0.255	/24 (256 addresses)	-
apps	192.168.1.0 - 192.168.1.255	/24 (256 addresses)	-

Click Next

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Create virtual network

Basics Security IP addresses **Tags** Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
Environment	Production	All resources selected
		All resources selected

Click Next

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Review + create

Create virtual network ...

Basics Security IP addresses Tags Review + create

[View automation template](#)

Basics

Subscription	Surf City Boats
Resource Group	ProdRG
Name	vNet-App
Region	East US

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

IP addresses

Address space	192.168.0.0/16 (65,536 addresses)
Subnet	default (192.168.0.0/24) (256 addresses)
- Private subnet	Enabled
Subnet	apps (192.168.1.0/24) (256 addresses)
- Private subnet	Enabled

Tags

Environment	Production
Environment	Production
Environment	Production
Environment	Production
Environment	Production

Click Create

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portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F93713e8c-2af6-4dde-be9b-1f3b21397873%2FresourceGroups%2FProdRG%2Fproviders%2FMicrosoft.Resources%2Fdeployments%2FvNet-App-1...

Microsoft Azure

Home > vNet-App-1726789727413 | Overview

Deployment

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Deployment in progress... Deployment to resource group 'ProdRG' is in progress.

Overview

Inputs

Outputs

Template

Deployment name : vNet-App-1726789727413

Subscription : Surf City Boats

Resource group : ProdRG

Start time : 9/19/2024 7:48:53 PM

Correlation ID : c5c767c1-c301-49ff-ac1f-c706f649d1c0

Deployment details

Resource	Type	Status	Operation details
There are no resources to display.			

Give feedback

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portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F93713e8c-2af6-4dde-be9b-1f3b21397873%2FresourceGroups%2FProdRG%2Fproviders%2FMicrosoft.Resources%2Fdeployments%2FvNet-App-1...

Microsoft Azure

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Deployment

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Deployment succeeded... Deployment 'vNet-App-1726789727413' to resource group 'ProdRG' was successful.

Go to resource Pin to dashboard

Your deployment is complete

Deployment name : vNet-App-1726789727413

Subscription : Surf City Boats

Resource group : ProdRG

Start time : 9/19/2024 7:48:53 PM

Correlation ID : c5c767c1-c301-49ff-ac1f-c706f649d1c0

Deployment details

Next steps

Go to resource

Give feedback

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Cost management

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portal.azure.com/#@surfcityboats.com/resource/subscriptions/93713e8c-2af6-4dde-be9b-1f3b21397873/resourcegroups/ProdRG/providers/Microsoft.Network/virtualNetworks/vNet-App/overview

Microsoft Azure

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vNet-App

Virtual network

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

- Address space
- Connected devices
- Subnets
- Baselining
- DDoS protection
- Firewall
- Microsoft Defender for Cloud
- Network manager
- DNS servers
- Peerings
- Service endpoints
- Private endpoints
- Properties
- Locks

Monitoring

Automation

Help

Essentials

Resource group (move) : ProdRG

Location (move) : East US

Subscription (move) : Surf_City_Boats

Subscription ID : 93713e8c-2af6-4dde-be9b-1f3b21397873

Address space : 192.168.0.0/16

DNS servers : Azure provided DNS service

Flow timeout : Configure

BGP community string : Configure

Virtual network ID : 6a0a0ee8-41f3-4aa5-babb-a46f42474b72

Tags (edit) : Environment : Production

Topology Properties Capabilities (5) Recommendations Tutorials

Click Address space

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

Not configured

Peerings

Seamlessly connect two or more virtual networks.

Not configured

Microsoft Defender for Cloud

Strengthen the security posture of your environment.

Not configured

Private endpoints

Privately access Azure services without sending traffic across internet.

Not configured

portal.azure.com/#@surfcityboats.com/resource/subscriptions/93713e8c-2af6-4dde-be9b-1f3b21397873/resourcegroups/ProdRG/providers/Microsoft.Network/virtualNetworks/vNet-App/addressSpace

Microsoft Azure

Home > vNet-App-1726789727413 | Overview > vNet-App

vNet-App | Address space

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Click Subnets

The address space for a virtual network is composed of one or more non-overlapping address ranges that are specified in CIDR notation. The address range you define can be public or private (RFC 1918). [Learn more](#)

Address space	Address range	Address count
192.168.0.0/16	192.168.0.0 - 192.168.255.255	65,536
Add additional address range		

Peered virtual network address space

Peering name	Peered to	Address space	Address range
No results.			

